

Yu-Jie Zhang

Postdoctoral Researcher
University of Washington

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EDUCATION

The University of Tokyo , Japan Ph.D., Complexity Science and Engineering	October 2021 - September 2024 Supervisor: Prof. Masashi Sugiyama
Nanjing University , China M.Sc., Computer Science and Technology	June 16, 2021 Supervisor: Prof. Zhi-Hua Zhou
Tongji University , China B.Sc., Electronic Science and Technology	July 01, 2018 Graduated with Honors

WORK EXPERIENCE & ACTIVITIES

University of Washington , US Postdoctoral Researcher	November 2025- Now Supervisor: Prof. Kevin Jamieson
RIKEN Center for Advanced Intelligence Project , Japan Postdoctoral Researcher	January 2025- October 2025 Supervisor: Prof. Masashi Sugiyama
The Institute for AI and Beyond , Japan Research Assistant	April 2022- September 2024

RESEARCH INTEREST

I am generally interested in exploring the theoretical foundations of machine learning and developing methods with sound theoretical guarantees. Currently, I focus on developing provably adaptive and reliable methods for *non-stationary and open-world environments*. My research topics include:

- **Sequential Decision-making in Non-stationary Environments:** We develop online learning methods that adapt promptly to non-stationary environment with regret guarantees
- *Key Words:* online optimization, *bandits*, *reinforcement learning*, *dynamic regret bound*.
- **Supervised Learning in Open-world Environments:** We develop reliable methods to learn with weak supervision and handle unknown classes with excess risk guarantees.
- *Key Words:* distribution shift, *weakly supervised learning*, *unknown classes*, *excess risk bound*.

PUBLICATIONS

Preprints

1. Soichiro Nishimori, [Yu-Jie Zhang](#), Thanawat Lodkaew, Masashi Sugiyama. On Symmetric Losses for Robust Policy Optimization with Noisy Preferences. *In submission*.

Conference Publications

1. Yan-Feng Xie, [Yu-Jie Zhang](#), Peng Zhao, and Zhi-Hua Zhou. Dynamic Regret via Discounted-to-Dynamic Reduction with Applications to Curved Losses and Adam Optimizer. **ICML 2026**.
2. [Yu-Jie Zhang](#), Sheng-An Xu, Peng Zhao, Masashi Sugiyama. Generalized Linear Bandits: Almost Optimal Regret with One-Pass Update. **NeurIPS 2025**.

3. Yu-Jie Zhang, Peng Zhao, and Masashi Sugiyama. Non-stationary Online Learning for Curved Losses: Improved Dynamic Regret via Mixability. **ICML 2025**.
4. Jing Wang, Yu-Jie Zhang, Peng Zhao, and Zhi-Hua Zhou. Heavy-Tailed Linear Bandits: Huber Regression with One-Pass Update. **ICML 2025**.
5. Yuting Tang, Yivan Zhang, Johannes Ackermann, Yu-Jie Zhang, Soichiro Nishimori, Masashi Sugiyama. Recursive Reward Aggregation. **RLC 2025**.
6. Long-Fei Li, Yu-Jie Zhang, Peng Zhao, and Zhi-Hua Zhou. Provably Efficient Reinforcement Learning with Multinomial Logit Function Approximation. **NeurIPS 2024**.
7. Yu-Yang Qian, Peng Zhao, Yu-Jie Zhang, Masashi Sugiyama, Zhi-Hua Zhou. Efficient Non-stationary Online Learning by Wavelets with Applications to Online Distribution Shift Adaptation. **ICML 2024**
8. Wei Wang, Takashi Ishida, Yu-Jie Zhang, Gang Niu, and Masashi Sugiyama. Learning with Complementary Labels Revisited: A Consistent Approach via Negative-Unlabeled Learning. **ICML 2024**.
9. Yu-Jie Zhang and Masashi Sugiyama. Online (Multinomial) Logistic Bandit: Improved Regret and Constant Computation Cost. **NeurIPS 2023 [Spotlight]**
10. Yu-Jie Zhang, Zhen-Yu Zhang, Peng Zhao, and Masashi Sugiyama. Adapting to Continuous Covariate Shift via Online Density Ratio Estimation. **NeurIPS 2023**.
11. Xin-Qiang Cai, Yu-Jie Zhang, Chao-Kai Chiang and Masashi Sugiyama. Imitation Learning from Vague Feedback. In Advances in Neural Information Processing Systems 36 **NeurIPS 2023**.
12. Yong Bai*, Yu-Jie Zhang*, Peng Zhao, Masashi Sugiyama, and Zhi-Hua Zhou. Adapting to Online Label Shift with Provable Guarantees. **NeurIPS 2023**. (* equal contribution)
13. Zhen-Yu Zhang, Yu-Yang Qian, Yu-Jie Zhang, Yuan Jiang, Zhi-Hua Zhou. Adaptive Learning for Weakly Labeled Streams. **KDD 2022**.
14. Yu-Jie Zhang, Yu-Hu Yan, Peng Zhao and Zhi-Hua Zhou. Towards Enabling Learnware to Handle Unseen Jobs. In Proceedings of the 35th AAAI Conference on Artificial Intelligence (**AAAI**), 2021.
15. Peng Zhao, Yu-Jie Zhang and Zhi-Hua Zhou. Exploratory Machine Learning with Unknown Unknowns. **AAAI 2021**.
16. Yu-Jie Zhang, Peng Zhao, Lanjihong Ma and Zhi-Hua Zhou. An Unbiased Risk Estimator for Learning with Augmented Classes. **NeurIPS 2020**.
17. Peng Zhao, Yu-Jie Zhang, Lijun Zhang and Zhi-Hua Zhou. Dynamic Regret of Convex and Smooth Functions. **NeurIPS 2020**.
18. Yu-Jie Zhang, Peng Zhao, and Zhi-Hua Zhou. A Simple Online Algorithm for Competing with Dynamic Comparators. **UAI 2020**.

Journal Publications

17. Sijia Chen, Yu-Jie Zhang, Wei-Wei Tu, Peng Zhao, and Lijun Zhang. Optimistic Online Mirror Descent for Bridging Stochastic and Adversarial Online Convex Optimization. Journal of Machine Learning Research (**JMLR**), 25(178):1–62, 2024.
18. Peng Zhao, Yu-Jie Zhang, Lijun Zhang, and Zhi-Hua Zhou. Adaptivity and Non-stationarity: Problem-dependent Dynamic Regret for Online Convex Optimization. Journal of Machine Learning Research (**JMLR**), 25(98):1–52, 2024.
19. Peng Zhao, Jia-Wei Shan, Yu-Jie Zhang and Zhi-Hua Zhou. Exploratory Machine Learning with Unknown Unknowns. Artificial Intelligence (**AIJ**), 327:104059, 2024.

AWARDS & HONORS

- Top Reviewer for NeurIPS, 2025
- Dean's Award for Outstanding Achievement (Doctoral Course), GSFS, UTokyo, 2025.
- AISTATS 2025 Best Reviewer, 2025
- Top Reviewer for NeurIPS, 2023
- Top Reviewer for UAI, 2023
- Top Reviewer for NeurIPS, 2022
- The University of Tokyo Fellowship, Tokyo, 2021
- Outstanding Master Dissertation Award by Jiangsu Computer Society, Nanjing, 2021
- Excellent Graduate of Nanjing University, Nanjing, 2021
- National Graduate Scholarship for Master Student, MOE of PRC, 2020

ACADEMIC SERVICE

- **Reviewer for Conference:** NeurIPS (2021-2025), ICML (2022-2025), ICLR (2022-2026), AISTATS (2021-2026), UAI (2022-2024), AAAI (2021, 2024), IJCAI (2020-2023), ECAI (2020).
- **Reviewer for Journal:** Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Frontiers of Computer Science.